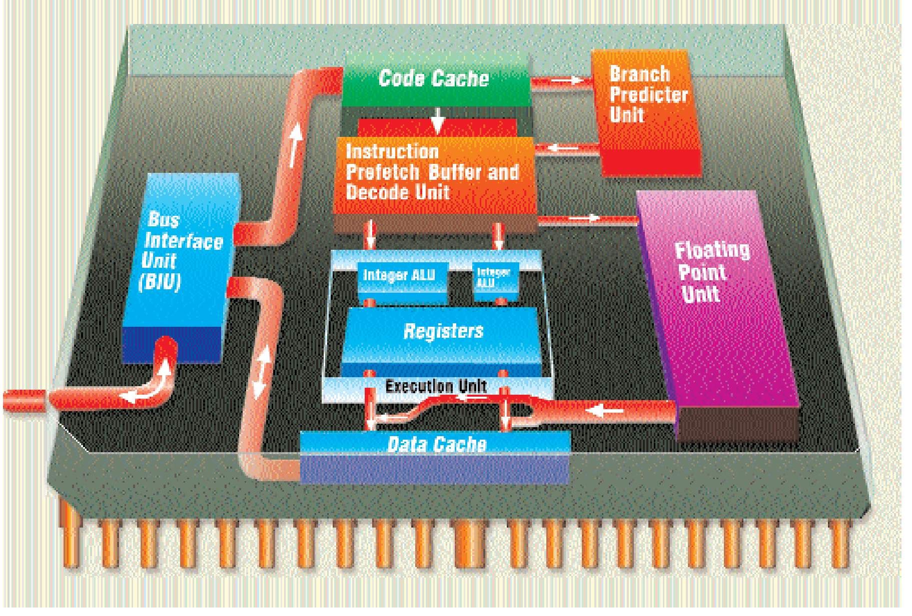
CPU (PROCESSORS)



Bus Interface Unit: It's responsible for fetching the required data to the processor.
Branch Predicter Unit: This block is responsible for predicting the future flow of the code and corresponding data. If this unit makes a mistake, the cached data has to be flushed, and new data has to be fetched—resulting in wastage of important CPU cycles.
Code Cache: This buffer area is used to store a pre-fetched code block.
Data Cache: The data required by the code is prefetched and stored in the data cache by the BIU and BPU.

Instruction Prefetch Buffer and Decode Unit: This block reads the instructions from the cache and executes them one after the other. If the instruction needs an integer operation, it is forwarded to the Integer ALU; if it requires a floating point operation, it is forwarded to the floating point unit.
Integer ALU: All integer operations are carried out here.
Floating Point Unit: All instructions dealing with floating points are carried out here.
Registers: These are used to store intermediate values during a block of operations.

Buying Tips

- ☑ Present-generation CPUs are power-hungry, so ensure that your power supply is rated at 350 W or more.
- ☑ Make sure you buy a genuine CPU. Look for the hologram seal on an Intel CPU box, or that the plastic casing on an AMD CPU is not broken.
- ☑ If you plan to do CPU-intensive tasks such as graphics processing, animation and gaming, opt for a CPU that has 256 KB or more of L2 cache. The Pentium 4 Northwood, for example, has 512 KB of L2 cache.
- ☑ The new Thoroughbred core AMD processors require less core voltage and have lower power dissipation, which makes them expensive. If all this doesn't really matter to you, opt for the cheaper Palomino cores.
- ☑ Invest in a good heat sink and fan combination for your processor; even if you do not plan to overclock your system, it will help keep things cool and stable.

Quick Find

Basic Use	YOU NEED	Decent performance in office productivity Applications
	LOOK FOR	Any processor with speeds in excess of 1 GHz
Inter Use	YOU NEED	Lag-free performance in graphic, multimedia and video oriented applications
	LOOK FOR	A processor with speeds in excess of 1.2 GHz
Prof Use	YOU NEED	Movies, music, gaming and complete home entertainment with oodles of processing power
	LOOK FOR	A processor with speeds in excess of 2 GHz

Facts

- A CPU contains many ground pins in order to reduce noise and prevent it from picking up stray voltages.
- According to Andy Moore, the founder of Intel, the number of capacitors on a chip would double every 18 months. Even though this statement is a couple of decades old, it still holds true.
- The 64-bit architecture that was initially meant only for servers and workstations is now powering desktop PCs as well.
- When the average temperature of a CPU rises by 10 degrees, its life-span is halved. This happens over a period of time due to a phenomenon called electromigration.

The Future

Initially, manufacturers were in a race to increase core clock and FSB speeds; next they raced to reach the 64-bit architecture for desktops. Now it's the turn of the software industry to rise to 64-bit operating systems, drivers, software, etc. Microsoft has already released a 64-bit version of its Windows XP operating system. It will take some time for software vendors to come up with software optimised for these processors—a time-consuming task, but one that is inevitable. There are other vendors such as Sun, IBM and HP that are in the race to keep their customer base intact by offering a 64-bit platform.